

2. Stable curves

Definition 2.1. A stable supercurve X/S is a proper supercurve over S , in the sense of 1.4, whose geometric fibres X_s have reductions

$$(X_s, \mathcal{O}_{X_s}^+)$$

which are stable curves.

For S the spectrum of an algebraically closed field k , it is easy, using 1.5, to determine, locally for the étale topology, the possible singularities of stable supercurves. On the singularity $xy = 0$, there are exactly two isomorphism types of coherent Cohen-Macaulay sheaves generically of rank 1. For C the curve, and B_1, B_2 its two branches, these are

$$\mathcal{O}_C \quad \text{and} \quad \mathcal{O}_{B_1} \oplus \mathcal{O}_{B_2}.$$

Both sheaves are isomorphic to their dual, with values in $\mathcal{O}_C$ or in ω_C , which is the same thing here. Moreover, every automorphism of either sheaf admits, locally for the étale topology, a square root, and 1.5 gives two possible types of supercurve singularities. Here is a description of the two possibilities by coordinates and equations. In each case I give the coordinates, the equations, a description of ω_X as a subsheaf of the direct image of its restriction to the smooth locus, and then the description of δ .

2.2. I. Coordinates z_1, z_2, θ ; equation

$$z_1 z_2 = 0.$$

The complement of the singular point is the union of two branches, with coordinates

$$\begin{cases} B_1 : z_1, \theta & (z_2 = 0), \\ B_2 : z_2, \theta & (z_1 = 0). \end{cases}$$

The sheaf ω_X is free with basis ℓ , given on the two branches by

$$\ell = \begin{cases} \begin{bmatrix} dz_1/z_1 \\ d\theta \end{bmatrix} & \text{on } B_1, \\ \begin{bmatrix} -dz_2/z_2 \\ d\theta \end{bmatrix} & \text{on } B_2. \end{cases}$$

The derivation δ is, on each branch,

$$f \mapsto \begin{cases} (\partial_\theta + \theta z_1 \partial_{z_1})f, \\ (\partial_\theta + \theta z_2 \partial_{z_2})f, \end{cases} \cdot \ell.$$

Let D denote the derivation given on the two branches by

$$\begin{cases} \partial_\theta + \theta z_1 \partial_{z_1} & (B_1), \\ \partial_\theta + \theta z_2 \partial_{z_2} & (B_2), \end{cases} \quad D : \mathcal{O} \rightarrow \mathcal{O} \text{ odd.}$$

II. Coordinates $z_1, z_2, \theta_1, \theta_2$; equations

$$z_1 z_2 = z_1 \theta_2 = \theta_1 z_2 = \theta_1 \theta_2 = 0.$$

The complement of the singular point is the union of two branches, with coordinates

$$z_1, \theta_1 \quad (z_2 = \theta_2 = 0), \quad z_2, \theta_2 \quad (z_1 = \theta_1 = 0).$$

The sheaf ω_X is generated by the following forms, given branchwise:

$$\left[\frac{dz_1}{d\theta_1} \right] := \begin{cases} \left[\frac{dz_1}{d\theta_1} \right], & \left[\frac{dz_2}{d\theta_2} \right] := \begin{cases} 0, \\ \left[\frac{dz_2}{d\theta_2} \right], \end{cases} \end{cases}$$

together with

$$\frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right] - \frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right] := \begin{cases} \frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right], \\ -\frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]. \end{cases}$$

On each branch,

$$\delta f = (\partial_{\theta_i} + \theta_i \partial_{z_i}) f \cdot \left[\frac{dz_i}{d\theta_i} \right].$$

Set

$$\ell := \left[\frac{dz_1}{d\theta_1} \right] + \left[\frac{dz_2}{d\theta_2} \right]$$

and let D denote the odd derivation, away from the singular point, given by

$$D := \partial_{\theta_i} + \theta_i \partial_{z_i} \quad \text{on each branch.}$$

It is a rational derivation: it does not send $\mathcal{O}$ to $\mathcal{O}$. One has

$$\theta_1 \mapsto \begin{cases} 1 & \text{on } B_1, \\ 0 & \text{on } B_2, \end{cases}$$

but

$$\delta = D\ell$$

does send $\mathcal{O}$ to ω_X .

In I and II, $\omega_{X/S}$ may be computed as in 1.6. Let α_i be sections of $\omega_{X/S}$ on B_i with the singular point removed, $i = 1, 2$. For the α_i to come from a section of ω_X , it is necessary and sufficient that, for every $f \in \mathcal{O}_X$, inducing f_i on the branch B_i , one have

$$\text{Res}_0(f_1 \alpha_1) + \text{Res}_0(f_2 \alpha_2) = 0.$$

Recall that

$$\text{Res}_0 \left(\frac{\theta}{z} \left[\frac{dz}{d\theta} \right] \right) = 1.$$

In case I, each closed branch B_i may be viewed as a curve equipped with a spin structure ramified at 0: a square root of

$$\omega_{C_i}(0),$$

where $C_i = B_{i,\text{red}}$, with a simple pole allowed at 0. One glues B_1 and B_2 along the integral leaf $z_i = 0$ of the distributions

$$\partial_\theta + \theta(z_i \partial_{z_i}).$$

In case II, each closed branch B_i may be viewed as a smooth supercurve; the two branches are glued at a point.

2.3. We shall have to consider the following deformations of the supercurves I and II. The deformations are over $k[[t]]$.

I. Coordinates z_1, z_2, θ ; equation

$$z_1 z_2 = t.$$

In the plane (z_1, z_2) , the derivation $z_1 \partial_{z_1} - z_2 \partial_{z_2}$ annihilates t . The complement of the singular point $(0, 0)$ in the fibre $t = 0$ is covered by the two charts

- (1) z_1 invertible, coordinates (z_1, θ) , $z_2 = t/z_1$,
- (2) z_2 invertible, coordinates (z_2, θ) , $z_1 = t/z_2$.

The sheaf $\omega_{X/S}$ is free with basis ℓ given by

$$\ell = \frac{[dz_1 \wedge dz_2/d\theta]}{d(z_1 z_2)} = \begin{cases} \left[\frac{dz_1/z_1}{d\theta} \right] & \text{in chart (1),} \\ \left[-\frac{dz_2/z_2}{d\theta} \right] & \text{in chart (2).} \end{cases}$$

The derivation δ is given in the two charts by

- (1) $(\partial_\theta + \theta z_1 \partial_{z_1}) \ell$,
- (2) $(\partial_\theta - \theta z_2 \partial_{z_2}) \ell$.

Since δ sends $\mathcal{O}$ to $\omega_{X/S}$ away from $(0, 0)$, depth implies that this is also true at $(0, 0)$. Flatness is clear: this is a relative complete intersection.

II. Coordinates $z_1, z_2, \theta_1, \theta_2$; equations

$$z_1 z_2 = -t^2, \quad z_1 \theta_2 = t \theta_1, \quad z_2 \theta_1 = -t \theta_2, \quad \theta_1 \theta_2 = 0.$$

The charts of $X - (0, 0)$ are

- (1) z_1 invertible, coordinates z_1, θ_1 , $z_2 = -t^2/z_1$, $\theta_2 = t \theta_1/z_1$,
- (2) z_2 invertible, coordinates z_2, θ_2 , $z_1 = -t^2/z_2$, $\theta_1 = -t \theta_2/z_2$.

If A is the coordinate ring of the total space, then

$$A^+ = k[z_1, z_2, \sqrt{z_1 z_2}].$$

One may view A^+ as the subring of $k[\sqrt{z_1}, \sqrt{z_2}]$ formed by sums of monomials of integral total degree. Through

$$\theta_1 \mapsto \sqrt{z_1}, \quad \theta_2 \mapsto \sqrt{z_2},$$

one may then identify A^- with the A^+ -module of sums of monomials of half-integral, non-integral, total degree. This makes it clear that $t = \sqrt{z_1 z_2}$ is not a zero divisor, and that X is flat over $k[[t]]$.

Here are sections of $\omega_{X/S}$ away from 0, written in charts (1) and (2):

$$\begin{cases} \left[\frac{dz_1}{d\theta_1} \right], \\ -\frac{t}{z_2} \left[\frac{dz_2}{d\theta_2} \right], \end{cases} \quad \begin{cases} \frac{t}{z_1} \left[\frac{dz_1}{d\theta_1} \right], \\ \left[\frac{dz_2}{d\theta_2} \right], \end{cases} \quad \begin{cases} \frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right], \\ -\frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]. \end{cases}$$

A depth argument shows that these sections extend uniquely to sections on X . They lift the generators of ω on the fibre $t = 0$. Since $\omega_{X/S}$ is flat over S and its formation is compatible with base change, they generate $\omega_{X/S}$.

Finally, δ is given in each chart by

$$\delta f = (\partial_{\theta_i} + \theta_i \partial_{z_i}) f \left[\frac{dz_i}{d\theta_i} \right].$$

The compatibility on the intersection of the charts is checked directly.

2.4. Remark. The morphism

$$\delta : \Omega_{X/S}^1 \longrightarrow \omega_{X/S}$$

is given by

$$\begin{aligned} dz_1 &\longmapsto \theta_1 \left[\frac{dz_1}{d\theta_1} \right], & d\theta_1 &\longmapsto \left[\frac{dz_1}{d\theta_1} \right], \\ dz_2 &\longmapsto \theta_2 \left[\frac{dz_2}{d\theta_2} \right], & d\theta_2 &\longmapsto \left[\frac{dz_2}{d\theta_2} \right]. \end{aligned}$$

Its image is generated by $[dz_1/d\theta_1]$ and $[dz_2/d\theta_2]$. Its cokernel has length 1, concentrated at $(0, 0)$. Indeed, if one multiplies

$$\frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right] = -\frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]$$

by θ_1, θ_2, z_1 , or z_2 , one obtains respectively

$$0, \quad 0, \quad \theta_1 \left[\frac{dz_1}{d\theta_1} \right], \quad \theta_2 \left[\frac{dz_2}{d\theta_2} \right].$$

Theorem 2.5. The deformation 2.3 is the completed versal deformation of the supercurve 2.2.

More precisely, one should complete 2.2 and 2.3 at the origin and take formal versal deformations. Of course, these are deformations as supercurves, not merely as superschemes.

Theorem 2.6.

- (i) A stable supercurve has no nontrivial infinitesimal automorphism.
- (ii) The stack of stable supercurves of genus $g > 2$ is a proper and smooth algebraic stack.
- (iii) It compactifies the stack of smooth supercurves of genus g , with a normal-crossings divisor at infinity.

To tell the truth, I have verified 2.6 only modulo the super-analogue of Artin's representability theorem and the super-analogue of Schlessinger's formal prorepresentability theorem.

Standard arguments reduce 2.6(i),(ii) to 2.6(i) and 2.5, together with verification of the valuative criterion of properness. For the latter, it is enough to consider only schemes as bases and to pass to the language of 1.5. Finally, to prove 2.5, it suffices to prove it to first order: if 2.3 over $k[t]/(t^2)$ is the only first-order deformation, then the tangent space to the versal deformation space has dimension $(1, 0)$, and every deformation over $k[[t]]$ extending the one over $k[t]/(t^2)$ is versal.

It was by examining the valuative criterion of properness that I came upon the deformations 2.3. From this point of view they are not surprising. The need for type II comes, for example, from the fact that if a curve of genus g degenerates into two curves of genera g_1 and g_2 attached at one point, then ω_{X_0} has no square root, being of odd degree on each component.

The divisors at infinity are characterized by the fact that, on each of them, one of the singular points admits, locally for the étale topology, an equation 2.2. Remark 2.4 provides, over each of them, a section, “the singular point”, whose image is locally the support of

$$\text{coker}(\delta : \Omega_{X/S}^1 \longrightarrow \omega_{X/S}).$$

These divisors at infinity are subvarieties of codimension $(1, 0)$.

I confess that for 2.5 I have only a disgusting proof: the sheaf of calculations that follows. It is somewhat rough, but in writing it up cleanly I would hardly do better.